

System Design Document

OutYah

Jamaica Outing Discovery Platform

Week 3 Deliverable (July 13 - July 17, 2026) - System Design

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Course: Special Projects

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Repository: https://github.com/KhriseanStewart/VTDI_Platform

Live app: <https://vtdi-platform.vercel.app>

1. Architecture Overview

OutYah uses a client-server architecture: a React SPA on Vercel communicates with Supabase (Auth, Postgres, Storage) and Google Maps Platform APIs.

Browser (React/Vite)

- | - AuthContext / DataContext / AppContext
- | - Public pages + Admin portal
- + -> Supabase Auth + REST (PostgREST) + Storage
- + -> Google Maps JS + Directions

2. Technology Selection

- Frontend: React 19, Vite, React Router, Lucide icons
- Backend-as-a-Service: Supabase (Postgres, Auth, RLS, Storage)
- Maps: Google Maps JavaScript API + Directions API
- Hosting: Vercel (SPA rewrites in vercel.json)
- Version control: GitHub

3. Database Design (ERD description)

Core entities and relationships:

- profiles (1:1 auth.users) - role user|admin
- places - venues with lat/lng, images, hours JSON, reviews JSON
- events - optional FK to places
- posts - Instagram-style media; optional FK to places
- post_comments - FK to posts
- favorites - composite PK (user_id, place_id)
- plan_stops - composite PK (user_id, place_id) + position

auth.users --1:1-- profiles

places --1:N-- events

places --1:N-- posts --1:N-- post_comments

users --N:M-- places (via favorites)

users --N:M-- places (via plan_stops ordered)

4. Security Design (RLS)

- Public SELECT on places, events, posts, comments
- Admin ALL on catalog tables via is_admin()
- Users manage only their favorites and plan_stops
- Storage bucket media: public read, admin write

5. Interface Design

Public shell: desktop sidebar + mobile bottom nav. Home uses a Feed/Map segmented control. Admin uses a separate layout without public bottom navigation. Empty states provide clear CTAs when catalogs are empty.

6. Key UML-style Flows

6.1 Discovery to Plan

1. User opens Feed, filters/search posts

2. Opens Venue Detail
3. Favorites and/or Add to outing
4. Planner shows ordered stops + route map
5. Get Directions opens Google Maps URL

6.2 Admin content publish

1. Admin signs in
2. profiles.role verified as admin
3. CRUD place/event/post in /admin
4. Public DataContext refresh shows content

7. Deployment Design

- Build: bun/vite -> dist/
- Vercel env: VITE_SUPABASE_URL, VITE_SUPABASE_ANON_KEY, VITE_GOOGLE_MAPS_API_KEY
- DB script: supabase/full_setup.sql (schema + optional seed)
- Client never receives service_role key